

Question 18**1 mark**

Identify the remainder when $P(x) = 3x^3 - x^2 + 1$ is divided by $(x - 2)$.

- a) -27
- b) -19
- c) 11
- d) 21

Question 19**1 mark**

Identify the logarithmic form of $2^x = \frac{1}{4}$.

- a) $\log_2 x = \frac{1}{4}$
- b) $\log_x 2 = \frac{1}{4}$
- c) $\log_2 \left(\frac{1}{4} \right) = x$
- d) $\log_x \left(\frac{1}{4} \right) = 2$

Question 20**1 mark**

Leah's Pizzeria offers 9 different pizza toppings. Identify the expression that represents the number of different types of pizzas, with 3 different toppings, that can be made.

- a) ${}_9C_3$
- b) ${}_9P_3$
- c) $\frac{9!}{3!}$
- d) $9!3!$

Question 21**1 mark**

Given $(5, -4)$ is a point on the graph of $y = f(x)$, identify the corresponding point on the graph of $y = \frac{1}{f(x)}$.

a) $\left(\frac{1}{5}, -4\right)$

b) $\left(5, -\frac{1}{4}\right)$

c) $\left(\frac{1}{5}, -\frac{1}{4}\right)$

d) $(-4, 5)$

Question 22**1 mark**

Identify the non-permissible value of x for $1 + \sec x$ over $[0, \pi]$.

a) 0

b) $\frac{\pi}{4}$

c) $\frac{\pi}{2}$

d) π

Question 23**1 mark**

Indicate the combination that represents the circled term in the given row of Pascal's triangle.

$$1 \quad 4 \quad 6 \quad \textcircled{4} \quad 1$$

a) ${}_4C_3$

b) ${}_4C_4$

c) ${}_5C_3$

d) ${}_5C_4$

Question 24**1 mark**

Identify the x -intercept on the graph of $f(x) = \sqrt{2(x+5)}$.

- a) -5
- b) 0
- c) $\sqrt{10}$
- d) 5

Question 25**1 mark**

Identify the coterminal angle of $\frac{\pi}{5}$ over the interval $-\pi \leq \theta \leq 4\pi$.

- a) $-\frac{9\pi}{5}$
- b) $-\frac{\pi}{5}$
- c) $\frac{3\pi}{5}$
- d) $\frac{11\pi}{5}$

Question 26**1 mark**

Given $f(x) = \{(2, 6), (3, 2), (3, 4), (6, 5)\}$, identify the value of $f(f(2))$.

- a) 3
- b) 4
- c) 5
- d) 6

Question 27

1 mark

The graph of $f(x) = (x - 1)^2$ is translated 2 units to the left and 3 units up. Identify the equation of the transformed graph, $g(x)$.

a) $g(x) = (x + 1)^2 + 3$

b) $g(x) = (x - 3)^2 + 3$

c) $g(x) = (x + 2)^2 + 3$

d) $g(x) = (x - 2)^2 + 3$

Question 28

2 marks 119

Given $\csc \theta = -\frac{8}{5}$, determine the exact value of $\cos 2\theta$.

Question 29

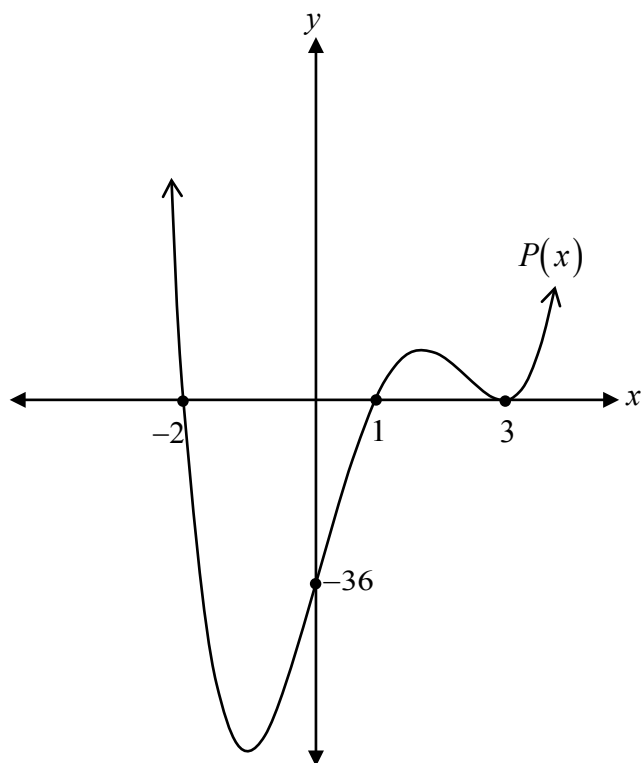
1 mark 120

Determine the period of the sinusoidal function, $f(x) = -6 \cos\left(\frac{\pi}{6}(x+1)\right) + 5$.

Question 30

2 marks 121

Determine, algebraically, the equation of $P(x)$, given the graph of the polynomial function $P(x)$.



$P(x) =$ _____

Question 31

4 marks 122

Solve $2\sin^2\theta - 7\sin\theta - 4 = 0$ where $\theta \in \mathbb{R}$.

Question 32

1 mark 123

Justify that the shapes of the graphs of $f(x) = (x+1)^2(x-1)$ and $g(x) = (x+1)^2(x-1)^3$ are different as they approach the x -intercept at $x=1$.

Question 33

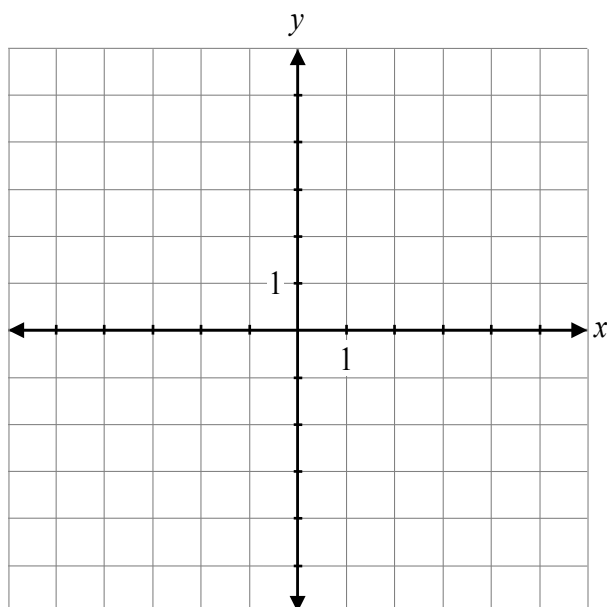
2 marks 124

Determine the exact value of $\cot \theta$ if $\cos \theta = -\frac{4}{7}$ and $\sin \theta$ is positive.

Question 34

3 marks 125

Sketch the graph of $f(x) = -\log_2(x) + 2$.



Question 35

1 mark 126

State the range of $f(x) = \sqrt{x+4}$.

Range: _____

Question 36

1 mark 127

Sophie correctly solved the logarithmic equation, $\log_7(x-1) = \log_7(2x-2)$.

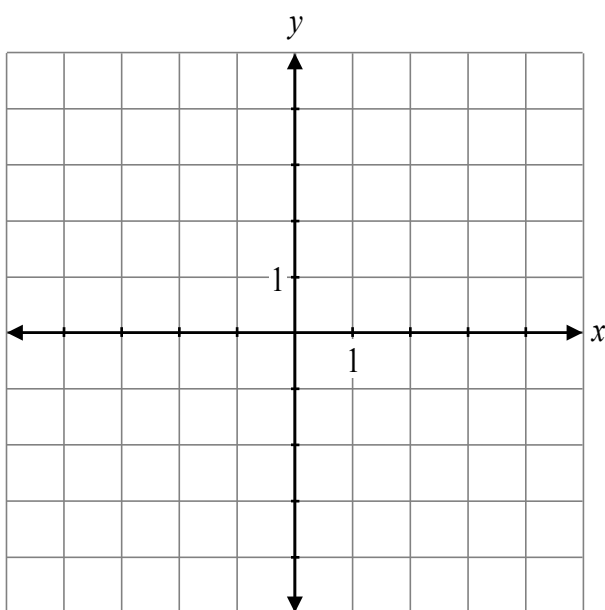
$$x-1 = 2x-2$$

$$-1+2 = 2x-x$$

$$~~1 = x~~$$

Explain why $x=1$ is an extraneous root.

Sketch the graph of $f(x) = \sqrt{4x} - 1$.



Question 38

3 marks 129

Solve, algebraically.

$${}_nC_2 = 2n + 7$$

Question 39

1 mark 130

Given $f(x) = x^2 - 1$ and $g(x) = x - 3$, explain why the domain of $h(x) = \frac{f(x)}{g(x)}$ has a restriction when $x = 3$.

Evaluate.

$$\frac{\cot\left(\frac{11\pi}{6}\right)\sin\left(-\frac{4\pi}{3}\right)}{\cos\left(\frac{2\pi}{3}\right)}$$

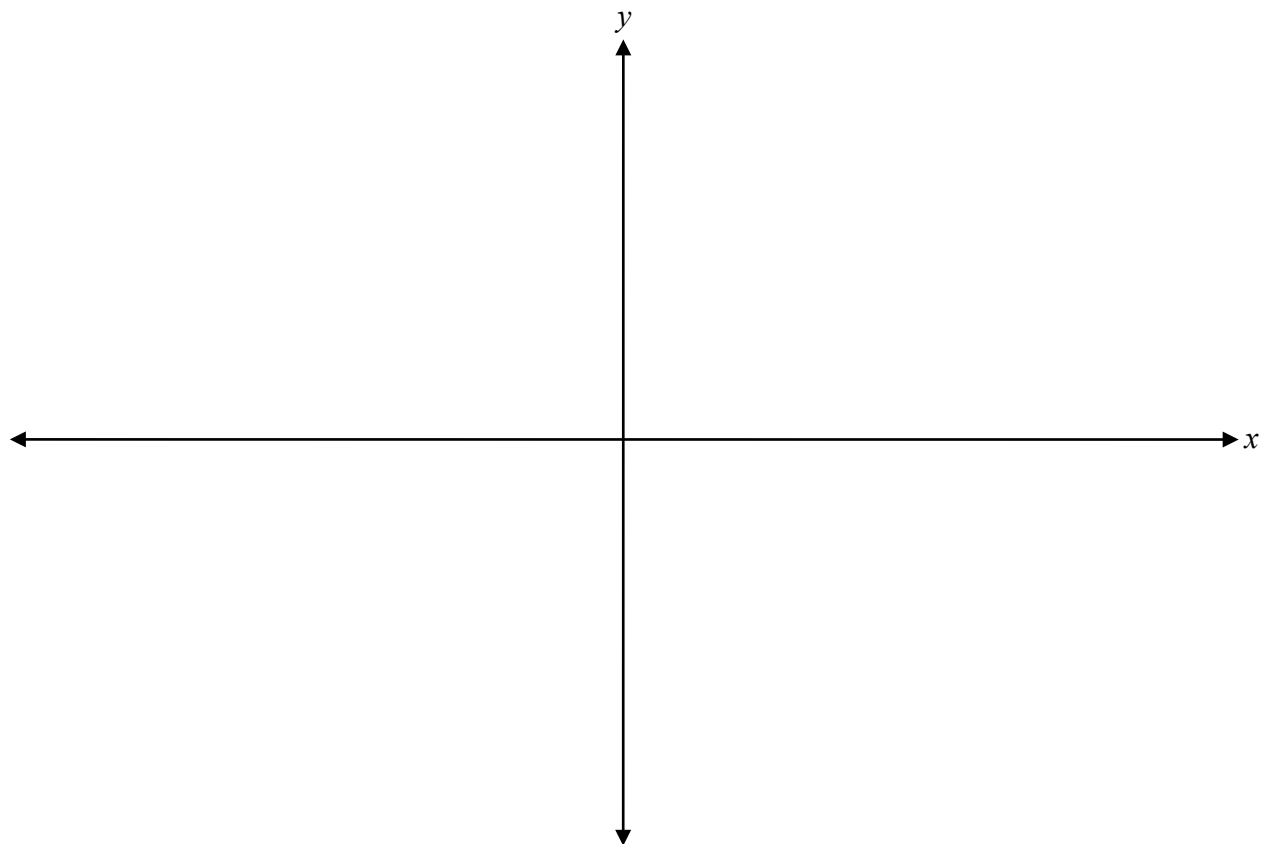
Question 41

1 mark 132

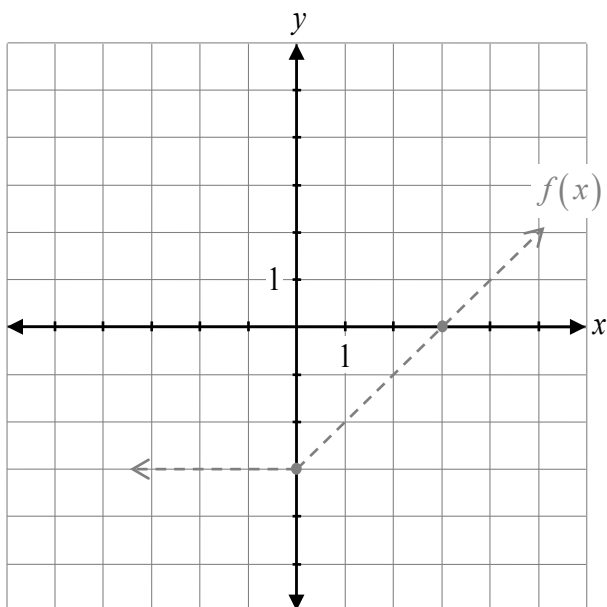
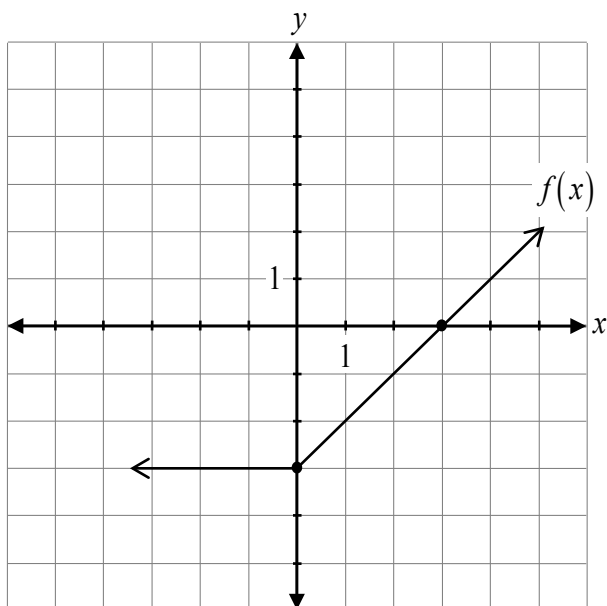
Solve, algebraically.

$$\log_2(\log_3 x) = 2$$

Sketch the graph of the function $y = 5 \sin\left(\frac{\pi}{4}x\right) + 1$ over the domain $[-4, 8]$.



Given the graph of $y = f(x)$, sketch the graph of $y = |f(-x)|$.

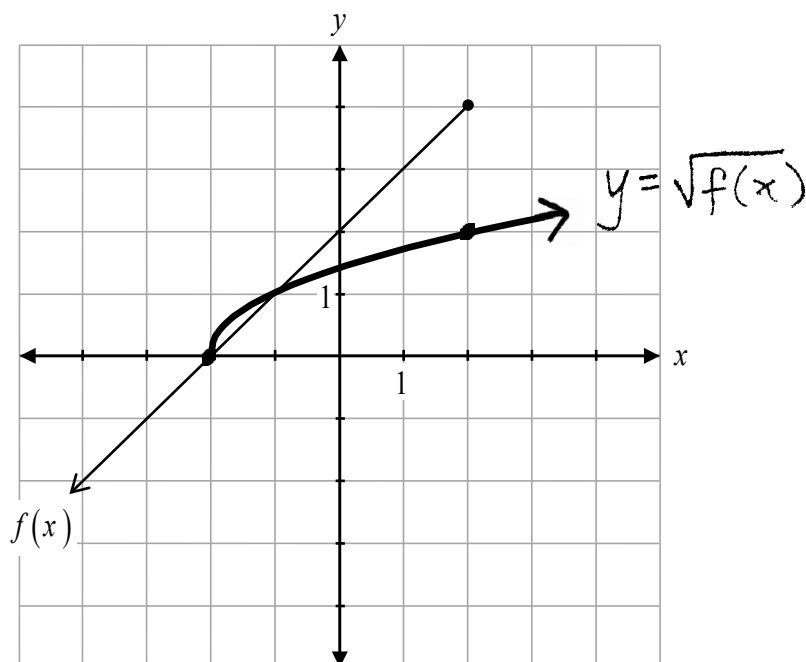


The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

Question 44

1 mark 135

Savannah used the graph of $y = f(x)$ to sketch the graph of $y = \sqrt{f(x)}$. Her solution is given below. Describe her error.



Determine the exact value of $\sin\left(\frac{13\pi}{12}\right)$.

Question 46

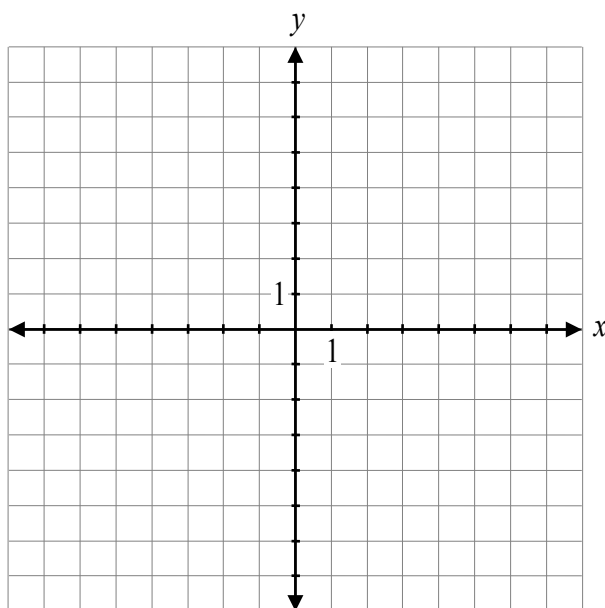
1 mark 137

State the equation of the horizontal asymptote of $f(x) = \frac{3x}{x-1}$.

Question 47

4 marks 138

Sketch the graph of $f(x) = \frac{5x - 10}{x^2 + x - 6}$.

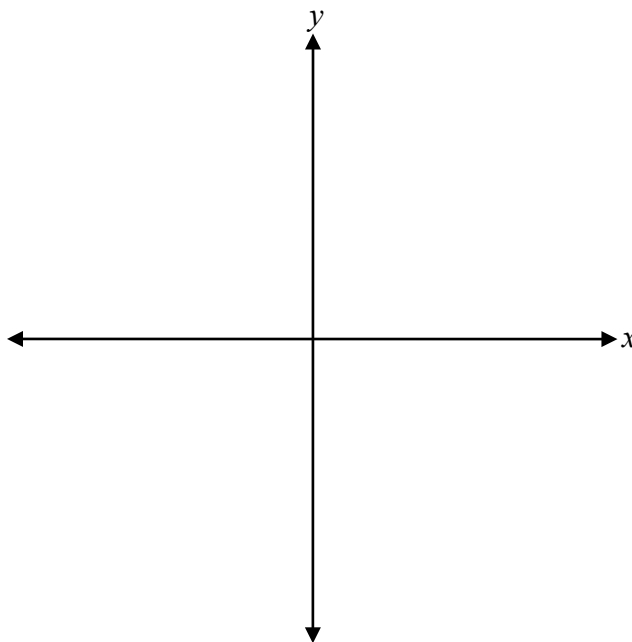


Question 48

2 marks 139

Determine, algebraically, the inverse of $f(x) = 3x + 4$.

Sketch the graph of $P(x) = -(x+1)(x-2)(x+3)$.



No marks will be awarded for work done on this page.

